

HemoLens Bill of Materials (BOM)

Designator	Component	Number of Units	Cost Per Unit [USD]	Total Cost [USD]	Source of Materials
Prusament PLA Galaxy Black	2258572	1	29.99	29.99	Prusa
Prusament PETG Galaxy Black	4715597	1	29.99	29.99	Prusa
M6 x 1 mm, 60mm long, Fully Threaded Hex Cap	91280A074	4	9.79	9.79	McMaster
Flanged, Oil-Embedded Sleeve Bearing for 6mm Shaft	6659K672	12	3.05	36.6	McMaster
50 mm MGN9 Linear Rail	B0D3T1MXKX	5	15.99	79.95	Amazon
150 mm MGN9 Linear Rail	B09SD9N2Q5	2		35.78	Amazon
M6 Hex Socket Set Screw Collar	6056N14	4	2.08	8.32	McMaster
M6 x 1 mm, Medium-Strength Steel Hex Nut	90592A016	6	5.21	5.21	McMaster
M3 x 0.5 mm, Low-Strength Steel Hex Nut	90591A250	18	2.81	2.81	McMaster
M3 x 0.5 mm, Medium-Strength Steel Thin Hex Nut	90695A033	2	4.23	4.23	McMaster

M3 x 0.5 mm, 8mm long, Fully Threaded Socket Cap	91290A113	12	12.52	12.52	McMaster
M3 x 0.5 mm, 6mm long, Fully Threaded Socket Cap	91290A111	30	11.32	11.32	McMaster
M3 x 0.5 mm, 4mm long, Fully Threaded Socket Cap	91290A109	36	13.48	13.48	McMaster
M8 x 1.25 mm, Medium- Strength Steel Hex Nut	90591A161	1	6.91	6.91	McMaster
M8 Clamping Two-Piece Shaft Collar	6063K14	1	9.93	9.93	McMaster
M8 x 1.25 mm, 130mm long, Fully Threaded Hex Cap	91287A463	1	3.50	3.50	McMaster
Small Circular Neodymium Magnets	B0CCXY6WBR	56	6.99	6.99	Amazon
100 mm MG9N Carriage and Rail	B09SD9GXVP	1	16.89	16.89	Amazon
Nema 17 Stepper Motor	B0CDRMHM82	1	15.98	15.98	Amazon
M3 Stainless Steel Ball Bearing	4668K216	2	24.14	48.28	McMaster
Arducam USB Camera	B096M5DKY6	1	49.99	49.99	Amazon

RECTANGULAR NEODYMIUM MAGNET					
M2 x 0.4 mm, Medium- Strength Steel Hex Nut	90591A265	4	2.22	2.22	McMaster
M2 x 0.4 mm, 8mm long, Fully Threaded Socket Cap	91290A015	4	18.76	18.76	McMaster
M12 Lens	B09W5KCNHK	1	16.00	16.00	Amazon
M6 Two-Piece Shaft Collar	6063K13	1	9.45	9.45	McMaster
M6 x 1 mm, 90mm long, Fully Threaded Hex Cap	91287A358	1	13.57	13.57	McMaster
M6 x 1 mm, Medium- Strength Steel Hex Nut	90591A151	1	3.14	3.14	McMaster
75 x 25 mm Fischerbrand Glass Microscope Slide	12-550-A3	1	79.00	79.00	Fischer
75 x 25 mm Rubber Gasket	B0B5V95MRG	1	9.99	9.99	Amazon
M3 x 0.5 mm Heat-Set Inserts for Plastic	94180A331	8	20.44	20.44	McMaster
M3 Stainless Steel Washer	93475A210	1	2.19	2.19	McMaster
M6 x 1 mm 100mm long, Fully Threaded Hex Cap	91287A361	2	14.31	14.31	McMaster

M6 x 1 mm Medium- Strength Steel Hex Nut	90591A151	2	3.14	3.14	McMaster
Plastic Barbed Tube Fitting	5117K85	2	14.69	14.69	McMaster
16 mm (Outer- Diameter Rubber) Grommet	B09XVDTQ8F	4	9.99	9.99	Amazon
ELEGOO Mega	B01H4ZDYCE	1	22.99	22.99	Amazon
YUNBO 12V 3A Power Supply	B07K225RR3	1	12.99	12.99	Amazon
DRV8825 and A4988 Stepper Motor Driver	B07YKH2JN8	1	6.99	6.99	Amazon
Wheatstone Amplifier Shield	RBC-Onl-38	1	48.16	48.16	RobotShop
L298N Motor Driver Board	B0CR6BX5QL	1	6.99	6.99	Amazon
DOMICA Small- Volume Aquarium Pump	B0892DKNR3	1	8.99	8.99	Amazon
Gikfun Mcro Peristaltic Pumps	B0C7KRJ9Z7	1	19.98	19.98	Amazon
Stepper Motor Cable	B0CMQN8L5R	1	8.99	8.99	Amazon
BARREL JACK	B089CXJPSC	1	23.49	23.49	Amazon
4 Pin Dupont Cable	B0789F523N	1	7.99	7.99	Amazon
Flow-Through Pressure Sensor	26PCDFG5G	2	89.11	178.22	Honeywell
Plastic Submersible Cord Grip	69915K11	2	3.85	7.7	McMaster

Quick and Secure Connect Terminal Block	8841T11	1	1.87	1.87	McMaster
HiLetgo 1602 LCD Keypad Shield	B00OGYXN8C	1	7.99	7.99	Amazon
1/16" ID Silicone Tubing	B0915Z455G	1	7.99	7.99	Amazon
IVYX Scientific Digital Water Bath	B07T81P5KN	1	49.99	49.99	Amazon